Building Technology

Engineering Mathematics I

Trainer Answer Guide - Keep Private

This document provides marking guidance. It is not linked from learner pages. Do not upload it publicly if answers should remain private.

General Marking Guidance

- Award method marks for correct formula selection.
- Award substitution marks where values are inserted correctly.
- Award accuracy marks for correct computation.
- Award communication marks for units and engineering interpretation.
- Accept alternative correct methods where the reasoning is valid.

Formula Guidance

Straight line: $y = mx + c$
$\tan \theta = \text{rise}/\text{run}$
Pyramid volume: $V = \frac{1}{3}Ah$
Circumference: $C = \pi d$
$dy/dx = f'(x)$

Expected Response Pattern

1. For questions involving roof pitch, the candidate should identify the quantities, choose a relevant formula, substitute correctly, compute accurately and interpret the result in building technology context.

2. For questions involving stair gradients, the candidate should identify the quantities, choose a relevant formula, substitute correctly, compute accurately and interpret the result in building technology context.

3. For questions involving excavation volumes, the candidate should identify the quantities, choose a relevant formula, substitute correctly, compute accurately and interpret the result in building technology context.

4. For questions involving cost curves, the candidate should identify the quantities, choose a relevant formula, substitute correctly, compute accurately and interpret the result in building technology context.

5. For questions involving building survey data, the candidate should identify the quantities, choose a relevant formula, substitute correctly, compute accurately and interpret the result in building technology context.

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.

2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.

3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.

4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.